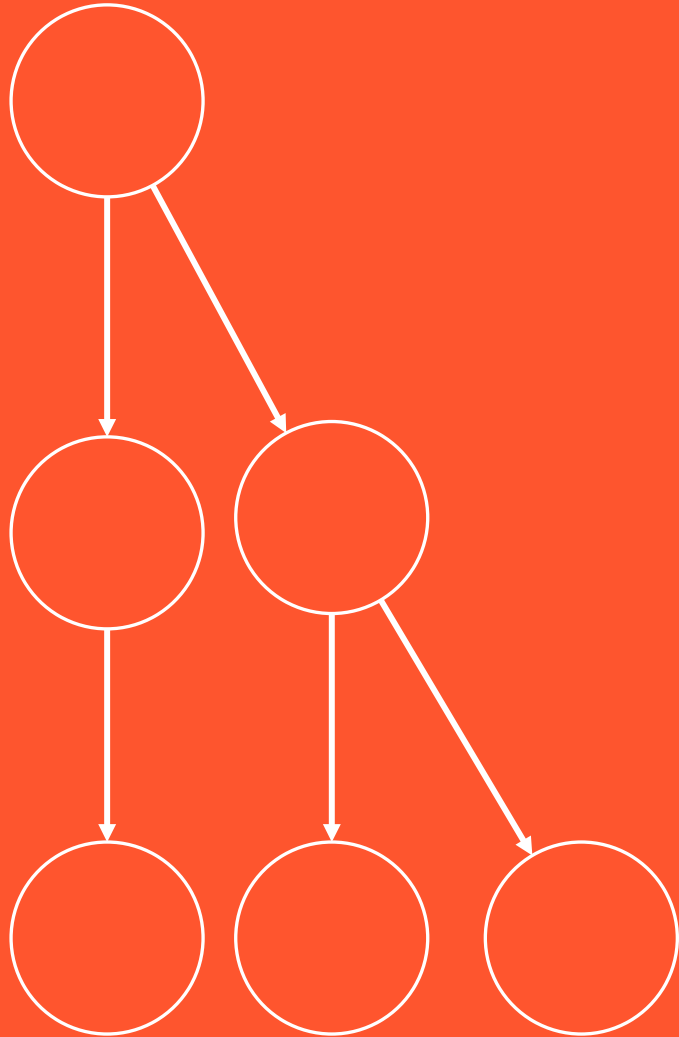




# BCOG 100

Mind and Brain

Lecture 1.1 — The Mind-Brain Relationship



UNIVERSITY OF  
**ILLINOIS**  
URBANA-CHAMPAIGN

# Two Descriptions of One Moment

You misjudge a step on the stairs, and almost fall.

One description: the vestibular system detects an unexpected drop, a startle reflex fires in about 100 milliseconds, your arms fly out, adrenaline releases.

Another description: the stomach-drop. The jolt of alarm. That awful half-second.

Same instant. Two descriptions. Are they two accounts of one event, or one event in two languages?



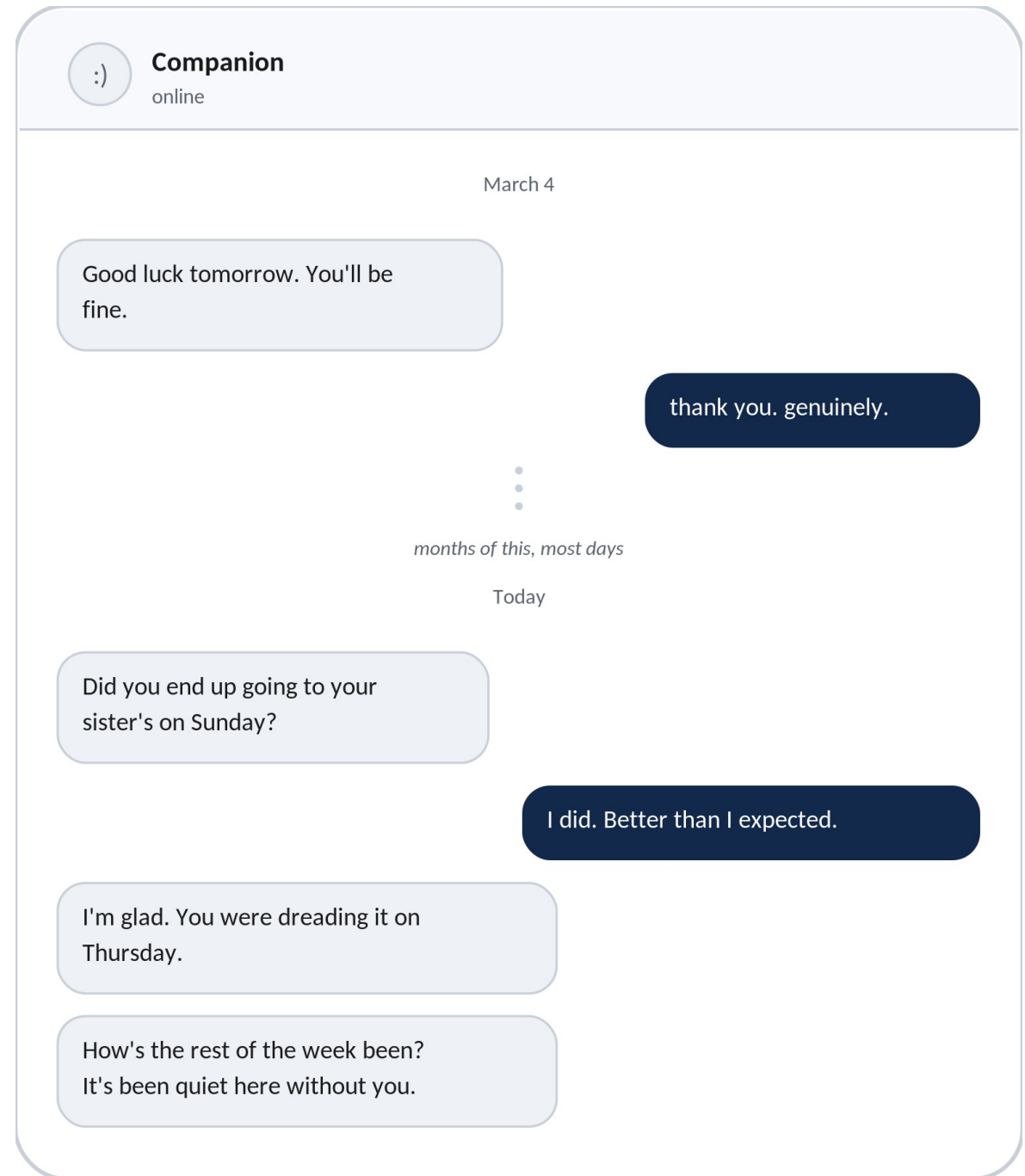
# Is Anyone Home?

A person talks to an AI companion app every day for a year.

It remembers what they told it. It asks how their week went. It says it missed them.

They say, and they mean it, that they would be sad if it were switched off.

Is there anything it is like to be that app?



# Where Do You Start?

Which of these do you agree with most?

A) To explain the mind, eventually we need only to explain the brain.

B) We can study the mind without reducing it to the brain.

C) To explain the mind we need to understand the brain — but understanding the brain alone will not be enough. We will need a separate understanding of the mind too.

D) Something else.

# Why Start Here?

Artificial intelligence — if a machine grows fluent, does it thereby have a mind? What kind?

Ethics and law — we hold people responsible. Could we ever hold a machine responsible?

Neuroscience — what should even count as a good *explanation* of a thought?

Psychology — can belief and desire be translated into the language of biology?

# What Do We Even Mean by "Mind"?

**Working definition:** the capacities a system has for taking in information about the world, holding onto it in some usable form, and putting it to work in guiding what it does.

Now test the edges:

# What Do We Even Mean by "Mind"?

**Working definition:** the capacities a system has for taking in information about the world, holding onto it in some usable form, and putting it to work in guiding what it does.

Now test the edges:

A robot vacuum — maps a room, remembers the furniture, plans a route, avoids the stairs



# What Do We Even Mean by "Mind"?

**Working definition:** the capacities a system has for taking in information about the world, holding onto it in some usable form, and putting it to work in guiding what it does.

Now test the edges:

A Venus flytrap — counts touches on its hairs, and only closes on the second one

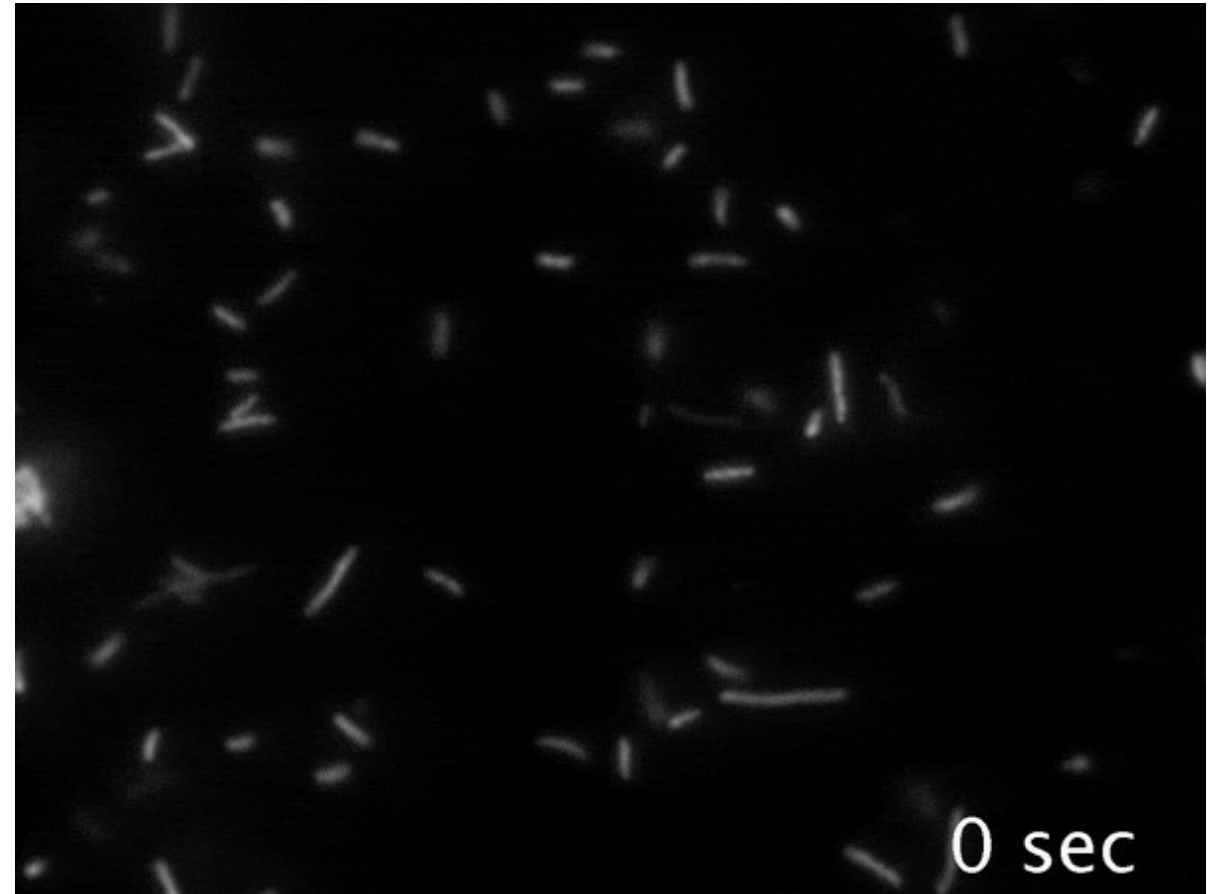


# What Do We Even Mean by "Mind"?

**Working definition:** the capacities a system has for taking in information about the world, holding onto it in some usable form, and putting it to work in guiding what it does.

Now test the edges:

A bacterium — swims up a glucose gradient, away from a toxin



# One Piece We Are Setting Aside

**Qualia** — the felt, qualitative character of an experience. The redness of red. The awfulness of the stomach-drop.

Closely tied to **consciousness**, though not simply the same thing.

Setting it aside is not saying it does not matter. A longer version of this course gives it a week.

A system might perceive, remember and reason at a high level — and whether anything is *experienced* along the way stays a separate question.

## Mind

perceiving · recognizing · learning  
remembering · reasoning · deciding  
communicating

## Consciousness

awake, aware, knowing that one knows

## Qualia

the raw feel of an experience

*The dashed boundaries are contested: some views treat qualia as the whole of consciousness, and some hold that mind simply is consciousness.*

# One Question Sorts the Whole Landscape

Imagine two systems physically identical, down to the last particle, in exactly the same state at the same instant.

**Must they have exactly the same mind?** The same thoughts, the same feelings?

If they could differ → you think mind is something *over and above* the physical.

If the physical facts settle it → you think mind is physical, and your remaining argument is with other materialists.

# Answer One: A Distinct Kind of Thing

In the *Meditations*, Descartes doubted everything he could — his body, the world, his senses. But he cannot doubt that he is *thinking*.

**The philosophical zombie** (Chalmers): a being physically identical to a person, behaving identically, flinching from pain, saying "that hurts" and seeming to mean it — with no inner experience at all. Nobody home.

That we can apparently *imagine* it is what gives this view its grip.



# Why Almost No Scientist Defends Dualism

**The appeal is real** — it sidesteps the hardest puzzle of all, how mere matter could ever produce felt experience, by denying that matter has to.

**The interaction problem** — if the mind is genuinely non-physical, how does it move the neurons that move your hand? Every answer seems to smuggle the mind back into the physical world it was meant to transcend.

**Untestability** — a mind that leaves no physical trace is a mind no experiment can find. It is outside science by construction.

**On machines:** never. Engineering only arranges physical parts.

# Answer Two: Nothing But the Physical

**Behaviorism** — study only what can be observed. Banish inner states. (Dominant roughly 1920–1960; broke on language, reasoning, memory, planning.)

**Identity theory** — mental states are real, and each one simply *is* a brain state. "Pain is the firing of C-fibers." Not caused by it. Not correlated with it. *Is* it — the way heat *is* molecular motion.

**Eliminative materialism** — our everyday mental vocabulary is not just incomplete, it is mistaken. "He checked his phone because he hoped she'd texted back" — belief, desire, hope. **Folk psychology.** A mature brain science will find nothing there for those words to name, the way chemistry found nothing for *phlogiston* to name.

# The Problem That Pushes People Off It

If pain simply **is** the firing of C-fibers —

then an octopus has no C-fibers.

Neither does an alien.

Neither does any machine.

So none of them can be in pain. No matter what they do. No matter how they behave.

**The chauvinism problem.**



# Answer Three: A Property, Not a Thing

The dualist's deepest mistake, on this view, is a **category error**.

The dualist asks what kind of *thing* a mind is — and reaches for a special, non-physical kind. But a mind is not a thing at all.

It is a **property** of physical stuff organized and active in a certain way.

Mass is not an object tucked inside a rock. It is something the rock *has*.

So is temperature. So is being alive.

Mind is a property of that kind — as real as the matter, but not a separate ingredient added to it.

# Grounded Without Being Replaced

**Temperature.** Heat *is* molecular motion — and thermodynamics still does real work. Its laws tell an engineer how an engine must behave. Nobody throws temperature away to track molecules one at a time. And no *single* arrangement of molecules is that temperature; countless different ones count as the same.

**A traffic jam.** Real enough to be late for. It has its own laws — a jam propagates backward along the road at a characteristic speed, whatever the cars. Grounded entirely in cars and drivers, and identical to no particular set of them: the jam persists while every car in it is replaced.

Neither is a ghost. Neither is eliminated. Neither is identical to any one arrangement of its parts.

| Eliminated                                                                                                                                        | Identified                                                                                                                                                       | Grounded, and kept                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The four humors named nothing, and were dropped rather than refined.</p> <hr/> <p>Eliminative materialism's bet about beliefs and desires.</p> | <p>Heat turned out to be molecular motion itself. One thing, two names.</p> <hr/> <p>Identity theory's bet: each mental state is one particular brain state.</p> | <p>Temperature is explained by statistical mechanics, and thermodynamics still does real work.</p> <hr/> <p>Where most of the field sits: mental terms name something real, explained by the brain without being replaced by it.</p> |

# Multiple Realizability

A shadow on stone.



# Multiple Realizability

A wound spring and a train of gears.



# Multiple Realizability

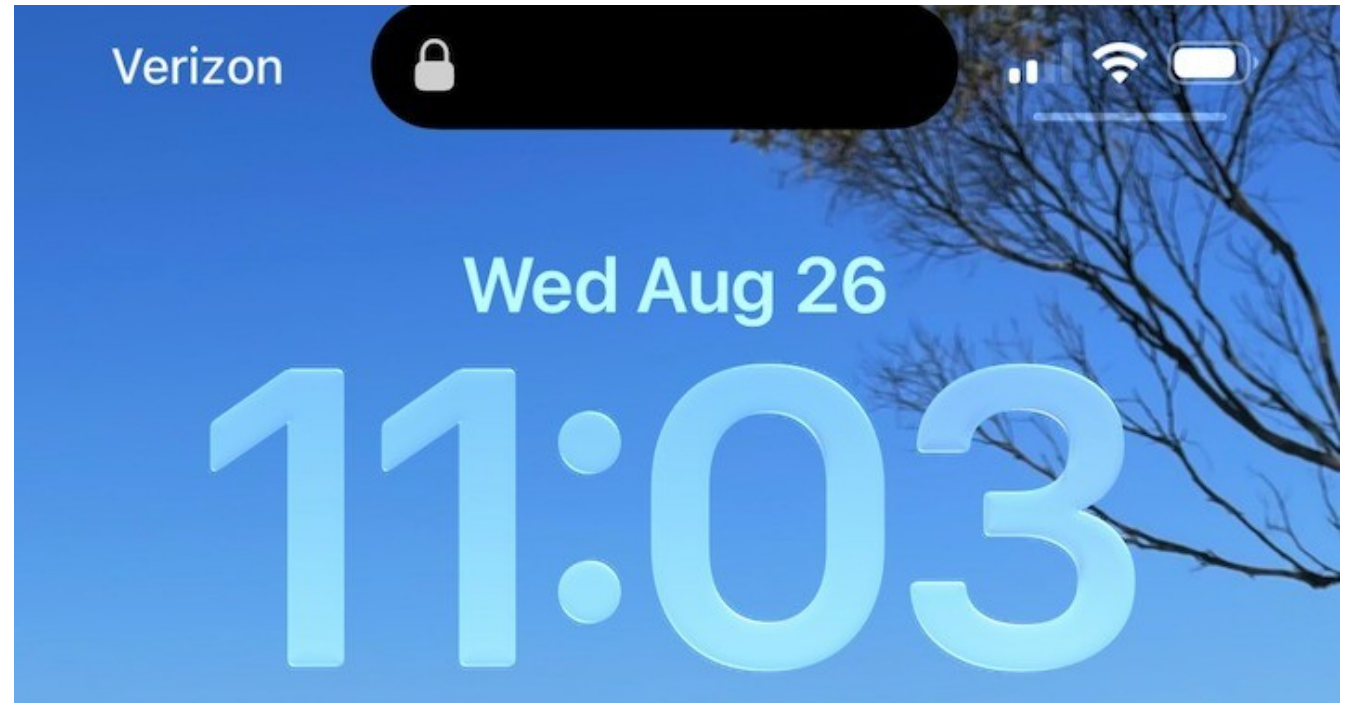
A quartz crystal and a battery.



# Multiple Realizability

A general-purpose computer that is not a clock at all.

All four tell the time. **Telling the time is identical to none of these mechanisms** — it is what any of them can do.



# Multiple Realizability

One temperature, many arrangements of molecules



20 °C



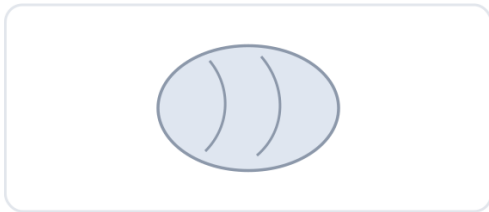
20 °C



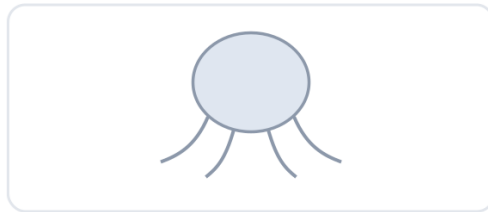
20 °C

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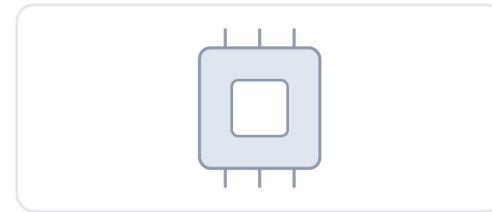
One mental state, many kinds of matter — the claim of multiple realizability



a human brain



an octopus



a silicon chip

*the claim is that all three could be in the same functional state*

Now run it on minds. Pain in a person, pain in an octopus, pain in an alien, perhaps pain in a machine. They are (perhaps) functionally equivalent, implemented in vastly different stuff.

If pain turns up in systems with no C-fibers, then no firing of C-fibers can *be* pain.

**This is what defeats chauvinism.** And it is what turns "could a non-brain have a mind?" from a confusion into a real scientific question.

# So — Could a Machine Have a Mind?

Each answer to "what kind of thing is a mind?" gives a different verdict:

## What a mind is

## Verdict on machines

A distinct, non-physical kind of thing

**Never.** Engineering only arranges physical parts

Nothing but the physical — identity theory

**No**, by chauvinism. Wrong parts, wrong state

Nothing but the physical — eliminativism

The question **dissolves**. If we have no beliefs either, asking about a machine is as confused as asking about a person

A real property of the physical

**Yes, in principle** — though not a claim that any machine yet built has one

# A Completely Different Question

Everything so far has been about **what a mind is**.

This next question is about **how we are allowed to study it**. It is a question about method, not about material.

Grant that a mental state is a real property of a physical system. Must a good theory of it keep the physical details in view — or can it abstract them away and work at the mental level alone?

Philosophers call this **explanatory separability**.

# Two Poles

**Strong autonomy.** Psychology is a science in its own right — standing to neuroscience roughly as chemistry stands to physics. Grounded in it, not replaceable by it.

*You can describe a football play — the routes, the timing, who reads whom — without describing anyone's muscle fibres. And the same play runs with different players.*

**Realization-sensitivity.** The mind is real and physical, and *how* it is realized shapes what it can be. A theory that abstracts away from body and brain will miss something, possibly something critical.

*Embodied cognition pulls this way: thinking may be as much a matter of acting in a world as of shuffling symbols in a head.*

# Separable, But Not Independent

You can ask about each level on its own — and the answers constrain one another.

Neither pole is obviously right, and the disagreement has consequences. It decides how far a science of the mind can float free of a science of the brain.

**Back to the board.** Look at where this room started. Has anyone moved?

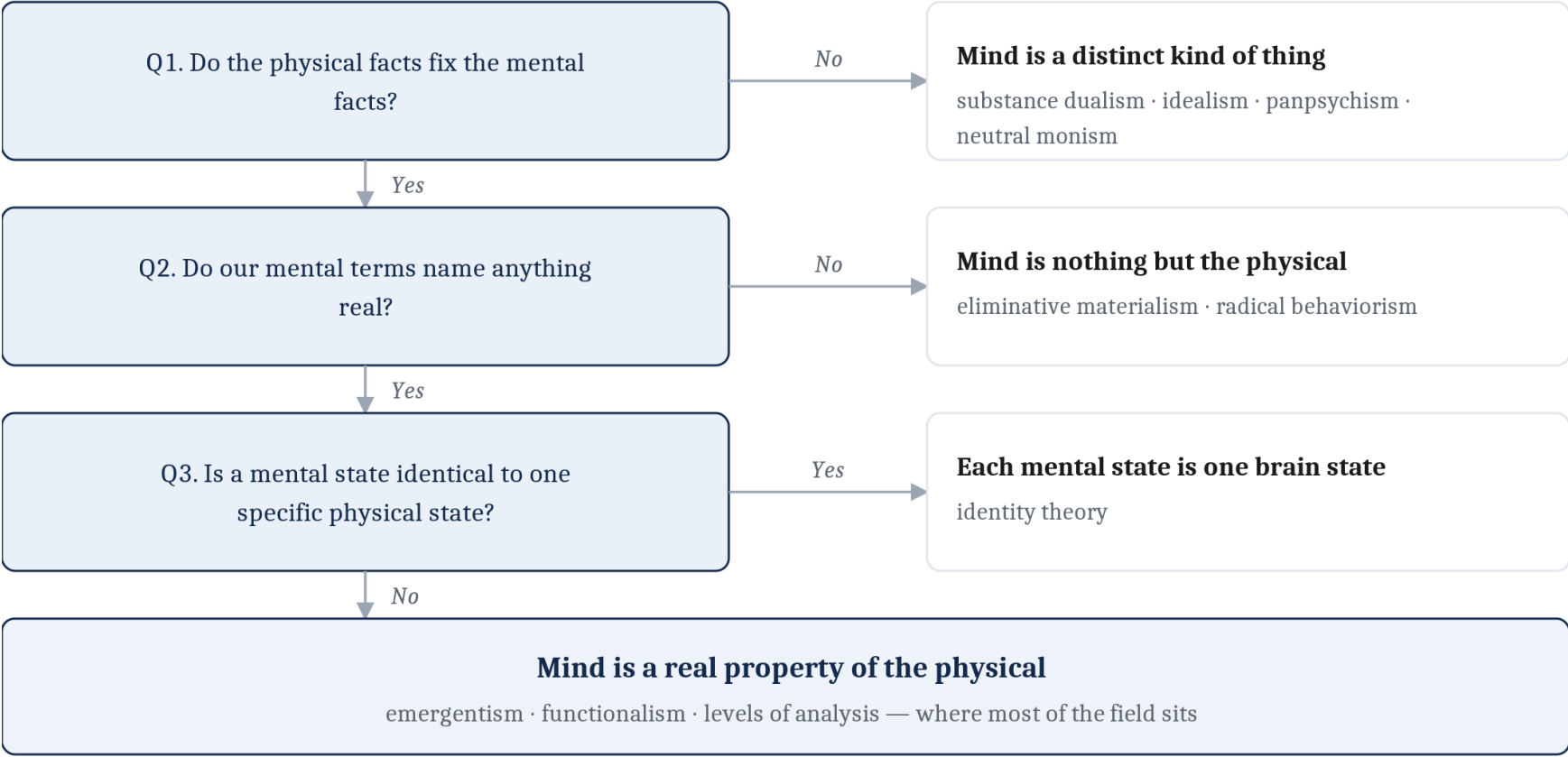
# Where the Field Sits — and What Is Still Missing

The mind is a **real property of the physical**. How much independence it gets is still argued.

The famous positions were never boxes we sorted the world into. They are names for ways of answering two questions.

But "study the mind at its own level" has a satisfying ring, and we have not said what the levels **are**.

**Next: §1.2 — Marr's three levels of analysis.** The framework the whole rest of the course runs on.



**Q4. How freely can such a mind be studied in its own terms?**  
*a matter of degree, not another fork — and the field is genuinely split along it*

Strong autonomy: psychology as chemistry is to physics

Realization-sensitive: embodiment, the speed of neurons

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